Medical Application and Healthcare
Bachelor of Artificial Intelligence - Academic Year 2025/2026
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Objective and Patient-Perceived Response to Anti-CGRP Treatment in Migraine: Mismatch and Factors

Introduction

Migraine is a chronic neurological disorder with a profound impact on patient quality of life. While anti-CGRP preventive therapies are highly effective in reducing attack frequency, clinical improvement in Monthly Migraine Days (MMDs) does not always correlate with a proportional reduction in patient-perceived disease burden. This discrepancy highlights the need to integrate objective clinical metrics with Patient-Reported Outcomes.
This project examines the relationship between objective and perceived responses to anti-CGRP treatment. **Task 1** analyzes the longitudinal association between MMDs and subjective outcomes (HIT-6, MIDAS, HADS), identifying individual-level discordance. **Task 2** explores whether baseline clinical or demographic factors mediate the gap between biological efficacy and patient experience.

Data and Study Design

The study integrates a baseline dataset (clinical/demographic traits) with a longitudinal dataset tracking patients across three treatment cycles, with visits a months 1, 3, 6, 9 and 12 per cycle.
The primary objective outcome is Monthly Migraine Days (MMDs). Subjective impact is assessed via the HIT-6 score, MIDAS scale and HADS (anxiety/depression). This multi-dimensional approach enables a comprehensive evaluation of treatment response by aligning biological efficacy with patient-reported outcomes.

Preprocessing and Missing Data

Before the main analyses, a preprocessing and missing data handling step was applied consistently across all analyses in the project. Baseline and longitudinal datasets were merged via `SUBJECT_ID`, with the time-related variables created to maintain the study's longitudinal structure. Given the heterogeneous missingness observed at baseline (Figure 1), complete-case analysis was ruled out to avoid significant sample size reduction.

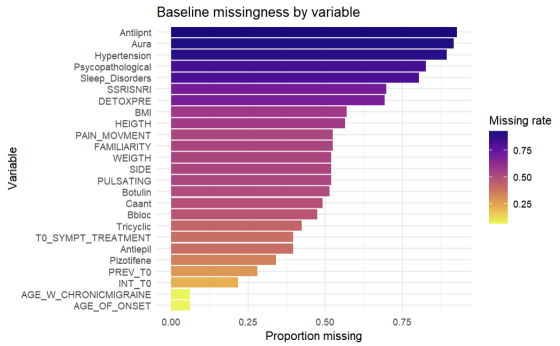

Figure 1: Baseline missingness by variable

Missing baseline covariates were handled using **Multiple Imputation by Chained Equations (MICE)** for variables with acceptable missingness levels. For longitudinal outcomes, missing values within each SUBJECT_ID x CYCLE to preserve the cyclical structure of the study and avoid information leakage across cycles. **Linear interpolation** was used in the primary analysis for continuous outcomes (e.g MMDs, HIT-6, HADS, INT), while a **Least Observation Carried Forward (LOCF)** strategy was used as a sensitivity analysis. Missing values occurring before the first observed visit within each cycle were not imputed. Two final analysis-ready datasets were generated (main and sensitivity) by merging the imputed baseline data with the longitudinal data via SUBJECT_ID.

Methods of Task 1

Longitudinal analysis using mixed-effects models

Linear mixed-effects models were fitted to evaluate changes in MMDs and HIT-6 scores over time. Treatment cycle and follow-up time (MONTH_num, coded as the scheduled visit months 1, 3, 6, 9, 12) were included as fixed effects, with a subject-specific random intercept to account for repeated measurements:

- **MMDs ~ CYCLE + MONTH_num + (1| SUBJECT_ID)**
- **HIT-6 ~ CYCLE + MONTH_num + (1| SUBJECT_ID)**

This approach allows estimation of the average changes while separating treatment-related effects from baseline differences between subjects. The same framework was exploratory applied to MIDAS and HADS scores.

Definition of response and discordance at Cycle 1

An individual-level analysis was conducted for Cycle 1 to define two response indicators:

- **Objective response**: ≥ 50% reduction in MMDs compared to cycle baseline.
- **Subjective response:** clinically meaningful improvement in HIT-6 score based on literature-consistent threshold.

End-of-cycle values were taken at Month 12 when available, otherwise, the last observed visit within Cycle 1 was used.

Patients were classified as concordant or discordant. The analysis focused on the primary discordant profile: patients achieving an objective response without a corresponding subjective improvement.

Sensitivity analysis

A sensitivity analysis was performed to evaluate the impact of the longitudinal missing-data strategy. All Task 1 analyses were repeated using a dataset imputed with Last Observation Carried Forward (LOCF) instead of linear interpolation. Model specifications, responder definitions, and discordance criteria were kept identical across analyses.

Results of Task 1

Longitudinal trends in objective and subjective outcomes

Figures 2 and 3 illustrate the clinical trajectories under anti-CGRP treatment. A consistent, progressive reduction in MMDs is evident across all cycles, signaling robust population-level efficacy in migraine frequency. Conversely, HIT-6 scores exhibit a more moderate and heterogeneous decline with higher variability. These trends suggest that objective physiological

improvements are more pronounced and stable than the corresponding changes in patient-perceived impact.

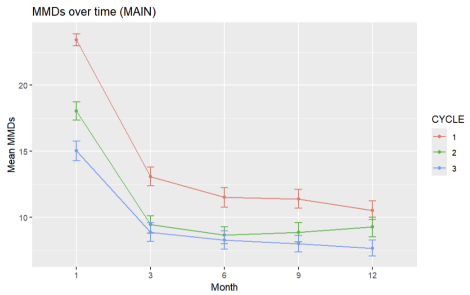

Figure 2: Longitudinal trends in MMDs

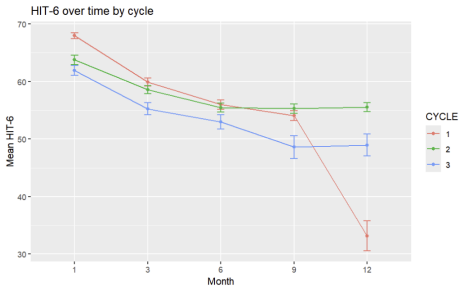

Figure 3: Longitudinal trends in HIT-6 scores

Mixed-effects model results

The linear mixed-effects models (*Table 1)* confirm the descriptive trends. For MMDs, the follow-up month shows a significant negative effect (Estimate ≈ -0.71, p < 0,001), validating a progressive reduction in migraine frequency. Significant negative coefficients for Cycle 2 and Cycle 3further support a cumulative treatment benefit over time.

effect	term	estimate	std.error	statistic	df	p.value	conf.low
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
fixed	(Intercept)	18.7627306	0.54834134	34.217246	250.585	2.008875e-96	17.6827855
fixed	CYCLE2	-1.2813919	0.38559705	-3.323137	1506.037	9.114631e-04	-2.0377561
fixed	CYCLE3	-2.2694055	0.40451261	-5.610222	1514.116	2.398691e-08	-3.0628699
fixed	MONTH_num	-0.7078243	0.03771568	-18.767372	1445.208	1.697175e-70	-0.7818076

For HIT-6, the effect of the follow-up month is also significantly negative (estimate ≈ -1.52, p < 0.001), but the cycle effects are less consistent (e.g., Cycle 3 is non-significant). These results confirm that while patient-perceived impact improves, the average benefit is less stable and follows a different trajectory compared to the objective reduction in migraine days.

effect	term	estimate	std.error	statistic	df	p.value	conf.low
<chr>	<chr>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>	<dbl>
fixed	(Intercept)	64.4723796	0.70615992	91.2999708	840.3579	0.000000e+00	63.086335
fixed	CYCLE2	3.0246391	0.71785158	4.2134603	1500.4687	2.664708e-05	1.616540
fixed	CYCLE3	-0.6310511	0.77214206	-0.8172733	1549.2507	4.138979e-01	-2.145605
fixed	MONTH_num	-1.5206299	0.07448936	-20.4140564	1448.4403	1.296753e-81	-1.666748

Objective-Subjective discordance at Cycle 1

The scatter plot in *Figure 4* visualizes the individual-level relationship between MMDs percentage change and HIT-6 score variations at the end of Cycle 1 (Month 12 or last available visit). While most patients follow a concordant path, the plot reveals significant inter-individual variability.

Clinical thresholds identify four response profiles, with particular interest in the primary discordant subgroup (upper-left quadrant): patient achieving a ≥ 50% reduction in migraine days without a corresponding clinically meaningful improvement in perceived impact (HIT-6). This mismatch highlights that biological efficacy does not always translate into subjective benefit during the initial treatment phase.

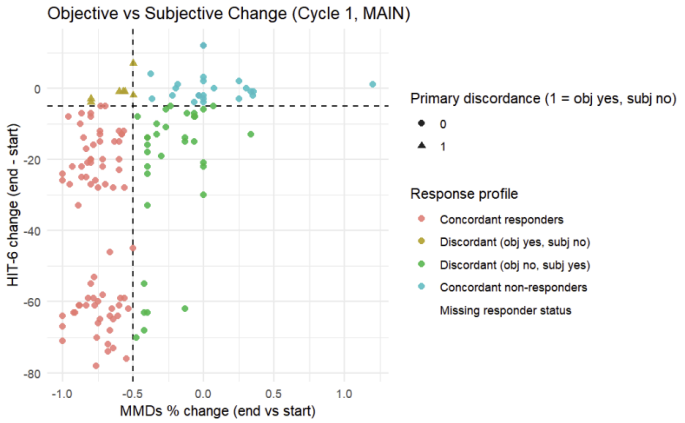

Figure 4: Objective vs subjective change at the end of Cycle 1

The concordance matrix (*Table 2*) confirms that this group accounts for 7 patients (5.0% of the sample). These results indicate that biological efficacy does not always translate into perceiving benefit. The presence of this mismatch, although numerically limited, underscores the importance of integrating Patient-Reported Outcomes into routine clinical evaluation to capture the full spectrum of treatment response.

obj_resp <int>	subj_resp <int>	n <int>	pct_total <dbl>
0	0	24	17.14286
0	1	34	24.28571
1	0	7	5.00000
1	1	75	53.57143

Results were qualitatively consistent under the LOCF sensitivity analysis, with similar longitudinal trends and discordance prevalence at Cycle 1, indicating that the observed mismatch is not driven by the missing-data strategy.

Methods of Task 2

Task 2 investigated whether baseline patient characteristics were associated with objective-subjective discordance observed at the end of Cycle 1. The outcome was the *Unhappy responder* profile (objective responder, but subjective non-responder, OBJ+/Subj-), defined in Task 1. Predictors included demographics (age, sex), clinical and comorbidity information (sleep disorders, migraine diagnosis) and baseline severity measures such as HADS scores, baseline disability (GGCEF_T0) and baseline pain intensity (INT_T0). Due to the limited number of discordant cases (n=7), associations were assessed using **univariate logistic regression models.** Results were reported as Odds Ratios (ORs) with 95% Confidence Intervals and interpreted as exploratory.

Results of Task 2

Baseline variables showed heterogeneous levels of missingness after data integration, with high missingness in several clinical and comorbidity-related variables, while demographic variables and core severity measures were largely complete (*Figure 5*). This limited the set of predictors that could be evaluated without excessive loss of sample size.

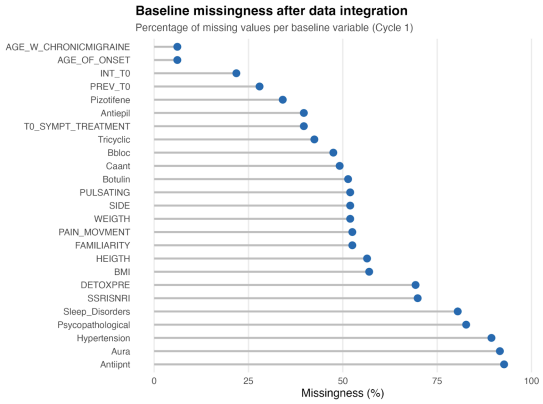

Figure 5: Baseline missingness after data integration

Univariable logistic regression (*Figure 6*) identified no statistically significant baseline predictors of the Unhappy responder profile at Cycle 1. Estimates were imprecise with wide confidence intervals due to the small sample size (n=7 events). The most notable trend was observed for baseline pain intensity (INT_T0), with an OR of 0.52 [0.24 - 1.08], suggesting a potential (though non-significant) association where higher pain intensity relates to lower odds of discordance. Overall, the lack of robust predictors suggests that the mismatch between objective and subjective benefit may depend on factors beyond baseline clinical measures.

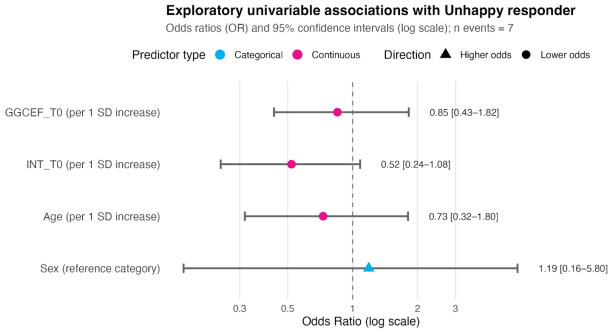

Figure 6: Exploratory univariable associations with objective-subjective discordance at Cycle 1

Discussion and Conclusion

This study investigated the alignment between objective efficacy and patient-perceived benefit of anti-CGRP treatment in a real-world setting. While longitudinal models confirm a significant, progressive reduction in migraine frequency, this improvement does not consistently translate into a proportional reduction in disease burden. The identification of a primary discordant subgroup (5.0%), patients with ≥ 50% MMD reduction but no meaningful HIT-6 improvement, highlights a critical gap between biological response and clinical experience.

Exploratory analyses did not identify robust baseline predictors of the Unhappy responder profile, suggesting that perceived benefit may be influenced by dynamic factors (e.g., psychosocial variables or long-term expectations) not fully captured at treatment onset. Despite limitations such as the observational design and baseline missingness, these findings underscore the necessity of a multi-dimensional approach. Integrating Patient-Reported Outcomes (PROs) and psychosocial assessments into routine clinical follow-up is essential to move beyond frequency-based metrics and achieve a truly patient-centered evaluation of migraine therapy.